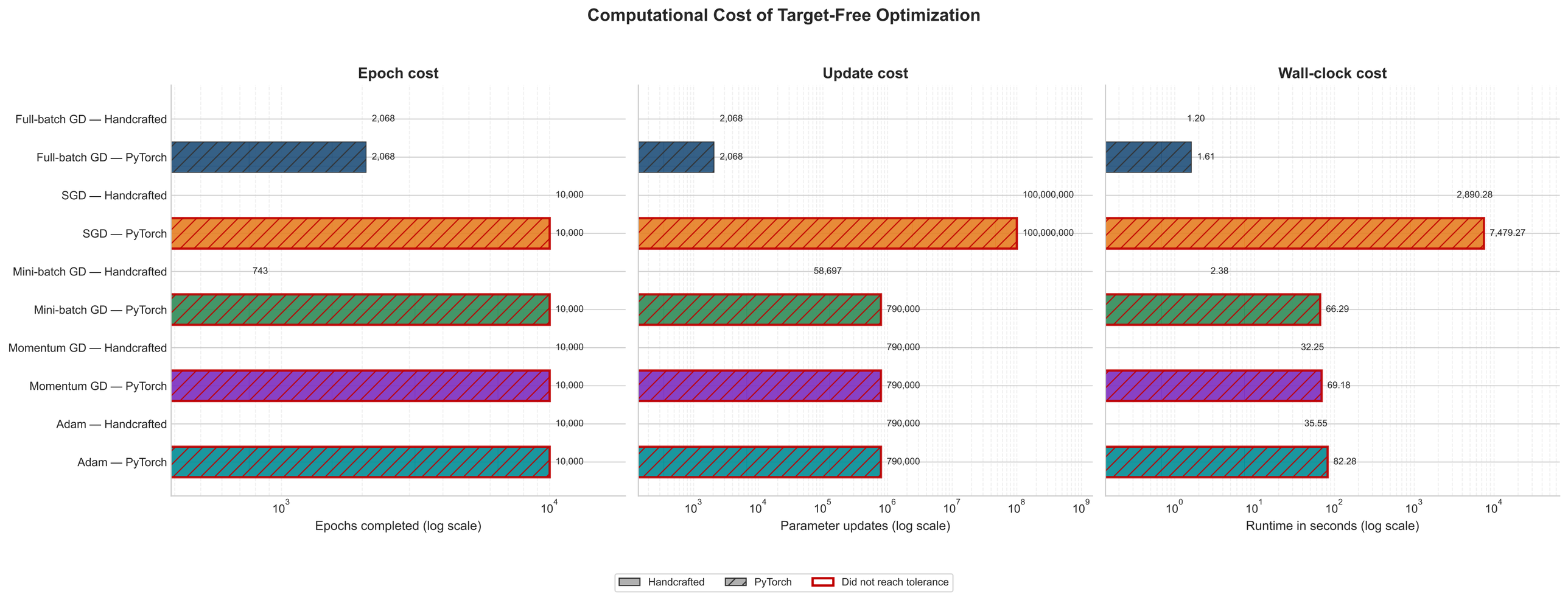
Computational Cost of Target-Free Optimization
Epoch cost
Update cost
Wall-clock cost
Full-batch GD — Handcrafted
Full-batch GD — PyTorch
SGD — Handcrafted
SGD — PyTorch
Mini-batch GD — Handcrafted
Mini-batch GD — PyTorch
Momentum GD — Handcrafted
Momentum GD — PyTorch
Adam — Handcrafted
Adam — PyTorch
2,068
2,068
10,000
10,000
743
10,000
10,000
10,000
10,000
2,068
2,068
100,000,000
100,000,000
58,697
790,000
790,000
790,000
790,000
1.20
1.61
2,890.28
7,479.27
2.38
66.29
32.25
69.18
35.55
82.28
$10^3$
$10^4$
Epochs completed (log scale)
$10^3$
$10^4$
$10^5$
$10^6$
$10^7$
$10^8$
$10^9$
Parameter updates (log scale)
$10^0$
$10^1$
$10^2$
$10^3$
$10^4$
Runtime in seconds (log scale)
Handcrafted
PyTorch
Did not reach tolerance